

Protein Structure Prediction of Hemoglobin Alpha Chain

Description

This project involved analyzing the predicted 3D structure of the human hemoglobin alpha chain using the AlphaFold Protein Structure Database to understand its structural and functional properties.

Method

- Retrieved protein structure from AlphaFold database
- Examined structural features and model confidence
- Analyzed predicted aligned error (PAE)

Results

The predicted structure showed a compact globular protein dominated by alpha-helices. Most regions exhibited high confidence scores (~99%), indicating a reliable structural model.

The PAE plot suggested strong structural consistency across the protein.

Caption

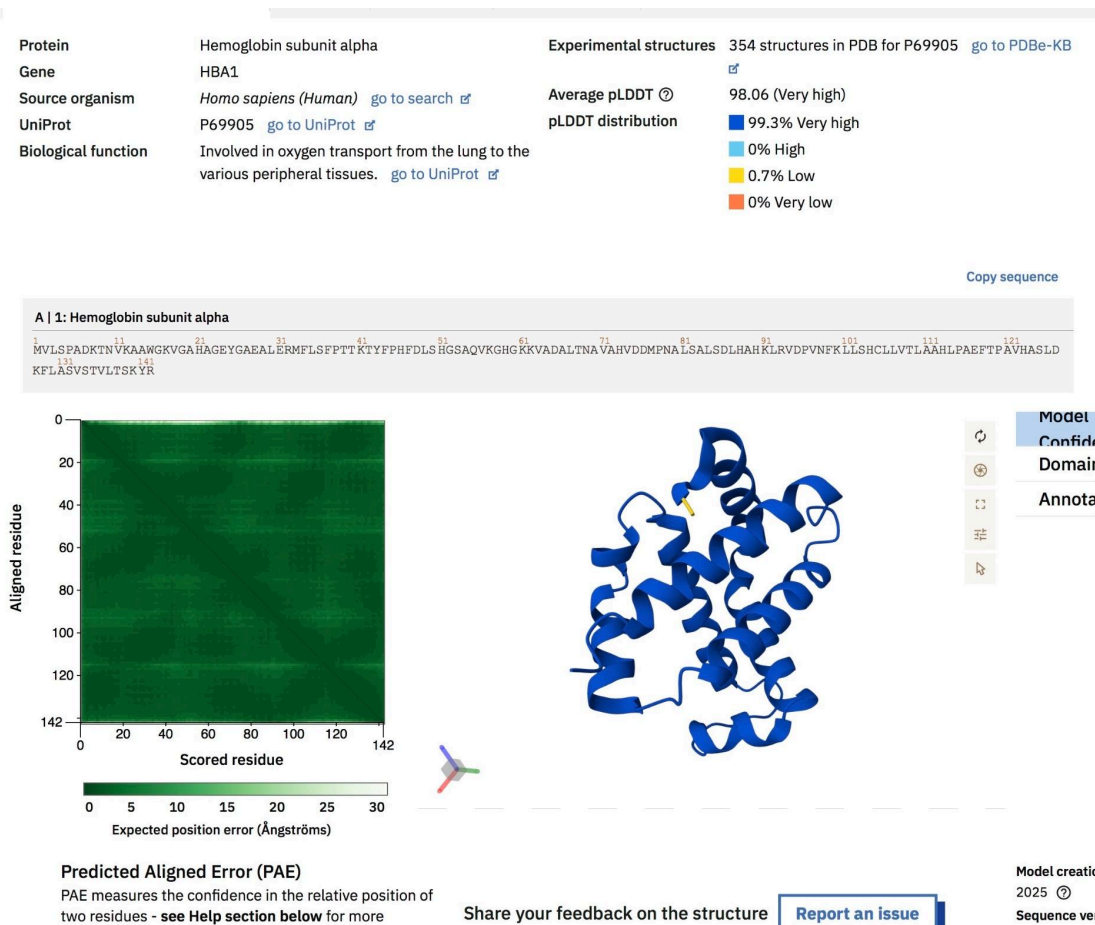


Figure 1: AlphaFold predicted structure of human hemoglobin alpha chain showing a highly confident alpha-helical structure and low predicted alignment error.

Interpretation

The alpha-helical structure is characteristic of globin proteins and supports their role in oxygen binding.

Skills Demonstrated

- Protein structure analysis
- Interpretation of confidence scores
- Understanding structure-function relationships
- Use of AlphaFold database